Week 8 — Activity: Energy Conservation and Geotherm Curvature

Geophysics 3A: Potential Fields & Geothermics

May 27, 2026

Purpose

To understand how crustal heat production alters lithospheric temperatures even when surface heat flow is unchanged.

Scenario

Two continental terranes have identical surface heat flow, q_0, but different amounts of heat supplied from the mantle at the base of the crust.

Tasks

1. Write the steady-state energy balance for the crust:

$$q_0 = q_b + \int_0^{z_c} A(z)\,dz$$

2. If Terrane A has a larger basal heat flow (q_b) than Terrane B, which must have higher integrated crustal heat production? Explain.
3. Consider the steady-state geotherm:

$$T(z) = T_0 + \frac{q_0}{k} z - \frac{A}{2k} z^2$$

   Identify which term controls curvature.
4. Without performing calculations, explain how greater curvature affects:
   - temperature at the Moho,
   - depth of intersection with the mantle adiabat.

Key Insight

Explain why surface gradients alone are insufficient to determine deep crustal temperatures.